from the atoms behind it. You watch as superhot atoms stream into the black hole from all sides. Once inside the hole, hotter than ever, they must vibrate faster than ever and radiate more strongly than ever, but their radiation cannot escape the hole's intense gravity. Nothing can escape. That is why the hole looks black; pitch-black.[4]

With your telescope, you examine the black sphere closely. It has an absolutely sharp edge, the hole's surface, the location of "no escape." Anything just *above* this surface, with sufficient effort, can escape from gravity's grip: A rocket can blast its way free; particles, if fired upward fast enough, can escape; light can escape. But just *below* the surface, gravity's grip is inexorable; nothing can ever escape from there, regardless of how hard it tries: not rockets, not particles, not light, not radiation of any sort; they can never reach your orbiting starship. The hole's surface, therefore, is like the horizon on Earth, beyond which you cannot see. That is why it has been named *the horizon of the black hole.*[5]

Your first mate, Kares, measures carefully the circumference of your starship's orbit. It is 1 million. kilometers, about half the circumference of the Moon's orbit around the Earth. She then looks out at the distant stars and watches them circle overhead as the ship moves. By timing their apparent motions, she infers that it takes 5 minutes and 46 seconds for the ship to encircle the hole once. This is the ship's *orbital period*

From the orbital period and circumference you can now compute the mass of the hole. Your method of computation is the same as was used by Isaac Newton in 1685 to compute the mass of the Sun: The more massive the object (Sun or hole), the stronger its gravitational pull, and therefore the faster must an orbiting body (planet or starship) move to avoid being sucked in, and thus the shorter the body's orbital period must be. By applying Newton's mathematical version of this gravitational law[6] to your ship's orbit, you compute that the black hole Hades has a mass ten times larger than that of the sun ("10 solar masses").[7]

You know that this hole was created long ago by the death of a star, a death in which the star, no longer able to resist the inward pull of its own gravity, imploded upon itself.[8] You also know that, when the star imploded, its mass did not change; the black hole Hades has the same mass today as its parent star had long ago—or almost the same. Hades' mass must actually